

Similarity statement

In the MP training set, there are in total 398 structures with PbX (X: Cl, Br, I), and 1627 structures with organics made of the elements C, H, N and O. Based on UMAP analysis, we observe that some have similarities in environments to the MP training set, but only 22 structures have a mixture of PbX and CHNO. From these, there are 14 3D HOIPs in which 12 of them have methylammonium (MA) as the organic cation, 3 0D HOIPs, and 4 cases of non-perovskite systems. In the MP, there is one 2D HOIP, which is the most similar to our dataset (mp-1194995), but this structure was not in the our dataset.

Performance summary

Stable NPT MD at ambient conditions for all 159 2D hybrid perovskite materials.

A.7 Protein Dynamics and Stability

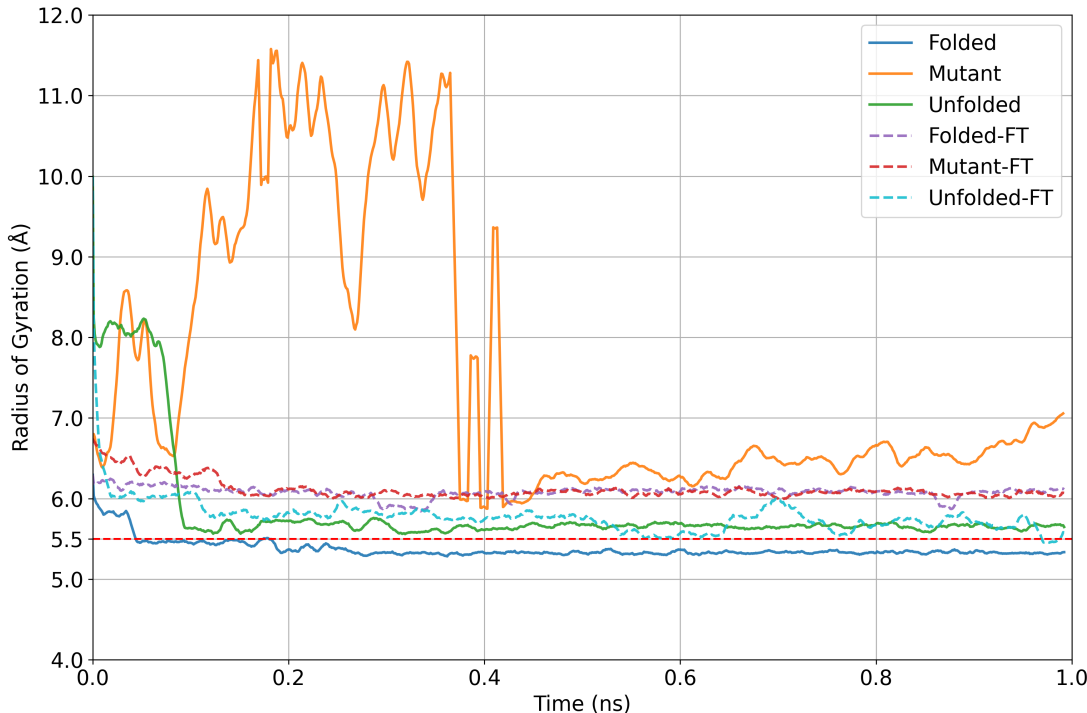


Figure S9: Plot of the radius of gyration ($\text{\AA}$) versus the time of simulation (ns) for the three simulations performed on Chignolin (PDB: 1UAO) and Chignolin mutant (PDB: 5AWL) with both MACE-MP-0b3 and the fine-tuned (FT) version. The radius of gyration curves are smoothed with a running mean over 10 steps to better visualise their overlap. The "Unfolded" simulation trajectories start from an unfolded structure of Chignolin, the "Mutant" lines correspond to the simulations for the Chignolin mutant and "Folded" corresponds to the simulations that start from a partially folded structure of Chignolin. The red line corresponds to the computed radius of gyration for the PDB 1UAO structure.

Understanding the dynamics of proteins is crucial for deciphering their biological function, and remains a core challenge of computational chemistry. Machine learning potentials have the capability of modelling non-covalent interactions, which are key components of secondary and tertiary structures of proteins, due to the quantum mechanical data on which they are trained on. In this section, we perform simulations on the well-known engineered Chignolin peptide (PDB: 1UAO), an artificial prototype for the protein folding phenomenon. This study diverges significantly from the majority of studies discussed in this manuscript as we are employing a machine-learning potential trained on inorganic crystals for a biological purpose.

We perform three separate simulations for each trained or fine-tuned model:

- A simulation starting from a partially folded structure of Chignolin.
- A simulation starting from an unfolded structure of Chignolin.
- A simulation of a mutant of Chignolin (PDB: 5AWL) starting from its PDB structure which contains 12 crystallographic waters.

Representative starting structures for the folded and unfolded Chignolin were obtained from Ref (195), specifically the first (folded) and last (unfolded) geometries from the 9543 conformations sampled by replica exchange molecular dynamics. We protonate negatively charged residues (1UAO: ASP3, GLU5, GLY10 | 5AWL: ASP3, GLU5, TYR10) and remove the proton on GLY1 for 1UAO and on TYR1 for 5AWL. We are aware that the real stable state of Chignolin is non-neutral and solvated but this test is performed in

a neutral environment as MACE-MP-0 was not trained with a charge-based loss and testing stability of the simulation is our primary goal. Simulations were performed at 300K for 1 ns at 1 fs time-step and a sampling frequency of 1 ps in the NVT ensemble using the Atomic Simulation Environment (ASE) package. The Langevin thermostat was employed with a friction coefficient of 10 ps⁻¹. MACE-MP-0b3 with the D3 dispersion correction was used for all simulations. Most simulations completed successfully in less than a day (elapsed-time) while one simulation ("Mutant") with the pre-trained MACE-MP-0b3 model took about three days.

We first analyse the results obtained from the pre-trained-only version of MACE-MP-0. From Figure S9 we can see that from its unfolded state (green line), Chignolin is quickly compacting at the start of the simulation and then maintains a stable radius of gyration, while the folding state (blue line) is maintained when starting from a partially folded state. Both simulations of Chignolin try to converge around the radius of gyration computed for the crystal structures 1UAO (red line), with the "Folded" simulation under-shooting and the "Unfolded" simulation slightly over-shooting it. On the other hand, the Chignolin mutant (yellow line) shows a perturbed radius of gyration due to some of the crystallographic waters losing contact to the protein early on in the simulation. Note that the radius of gyration was computed considering all the system's atoms. This was done to highlight any lack of stability in the simulation.

The simulations pursued with the fine-tuned (FT) model show improvements in terms of stability for the Chignolin Mutant (red line) as the crystallographic waters interact with the protein at any given time during the simulation. The unfolded Chignolin simulation (light-blue line) has an even faster formation of a compact protein structure compared to the pre-trained-only model, while the folded Chignolin simulation (purple line) is stable but has a shift towards a higher-average radius of gyration. To investigate this further, we performed a proton transfer analysis, shown in Figure S10.

On the left, we estimate the amount of proton transfers happening across time steps of each simulation. We use a simple distance-based detection to define the transfer of a proton. At every time step, we keep track of the closest heavy-atom neighbour of each proton. If in a time-step the proton has a new heavy atom neighbour with respect to the previous time-step and its distance to it is less or equal to 1.2 Å (resembling the fact that a bond has likely formed), then we increase the total transfer count by one. We observe a large number of proton transfers for the folded Chignolin and Chignolin mutant in the pre-trained MACE-MP-0, while none in the unfolded. The number of proton transfers drops drastically in the fine-tuned version of the model. In theory, we expect the transfers to be low as the system is neutralised and there are little to no water solvation shell with which the system could interact. Thus, it is clear that the fine-tuned model is closer to our expectations. This however might also mean that the fine-tuned model is biased towards weaker covalent and non-covalent interactions, which could also explain the higher radius of gyration observed in Figure S9 for all of the fine-tuned simulations.

We want to point out that some proton transfers are still observed in the Chignolin mutant even after fine-tuning. The proton transfer encountered happens between the neutralised COOH and NH₂ functional groups of GLY1 and GLY10 of Chignolin. This is a typical case of acid-base chemistry where we would expect the proton to transfer from the more acidic carboxylic acid group to the basic amine group, forming a zwitterion in physiological conditions (i.e. neutral pH). Overall this small case-study shows the capability of MACE in modelling reactivity in molecular dynamics simulations. We point out that these observed processes of proton transfer cannot be observed with classical force fields as they are generally not parametrised for such effects. Moreover, we conclude that fine-tuning the pre-trained foundation model is overall necessary to find a better agreement with the experiment and realistic physicochemical behaviour. However, more real-world tests, which are out of the scope of this section, are needed to confirm that the model behaves correctly. Such tests would include a fully solvated system in physiological pH conditions and use of a better-tailored fine-tuning dataset.

Similarity statement

The MP dataset encompasses only 99 structures exclusively composed of the elements hydrogen (H), carbon (C), oxygen (O), and nitrogen (N). Based on the UMAP analysis, the atomic environments of Chignolin and Chignolin mutant are clustered similarly to those of the filtered 99 structures (see Figure S11), which could mean that the environments of the training set are fairly similar to those of the test structures. However, after manual inspection, we observe that the filtered structures only resemble the protein under

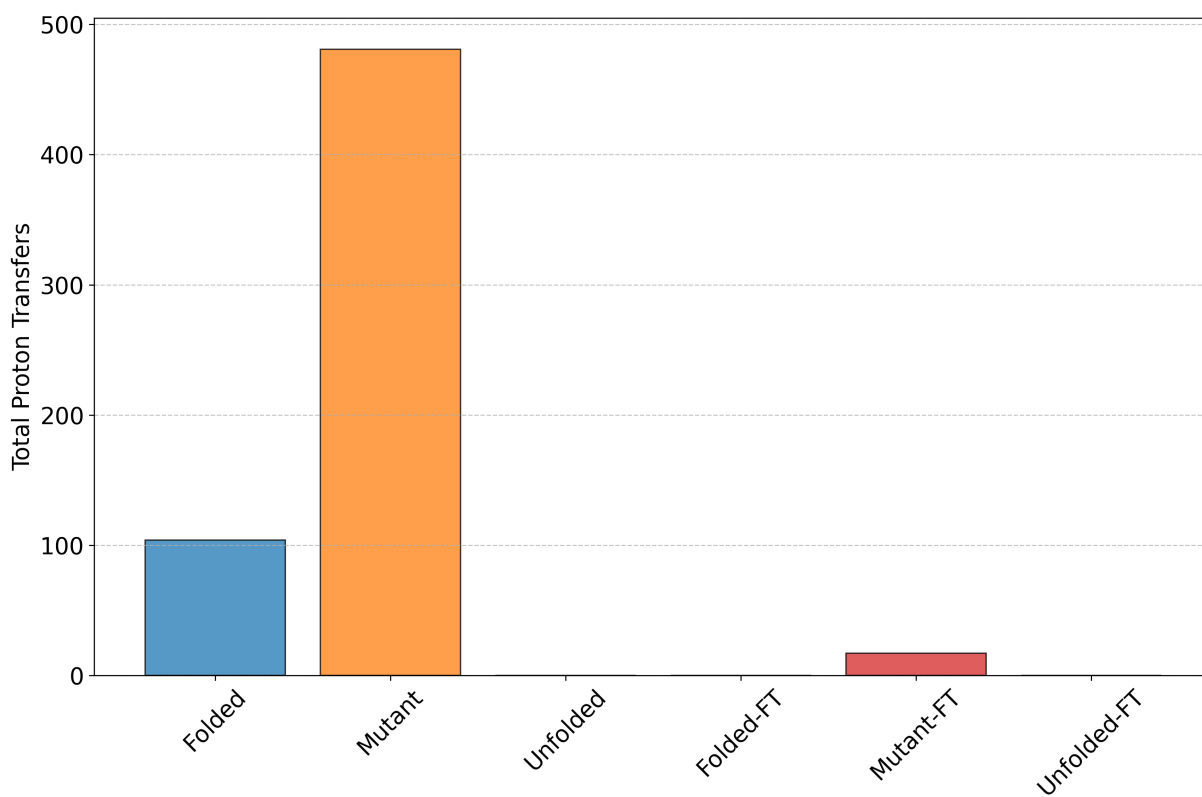


Figure S10: Histogram of protons transfers occurring during the 1 ns simulations. Total count of proton transfers throughout the simulation for each system.

study by some functional groups such as carboxylic acid ($-\text{COOH}$), amino groups ($-\text{NH}_2$), aromatic and amide groups. Some of the most similar structures are *e.g.* mp-998880, mp-1203308, mp-556151, mp-707289 and mp-1203544.

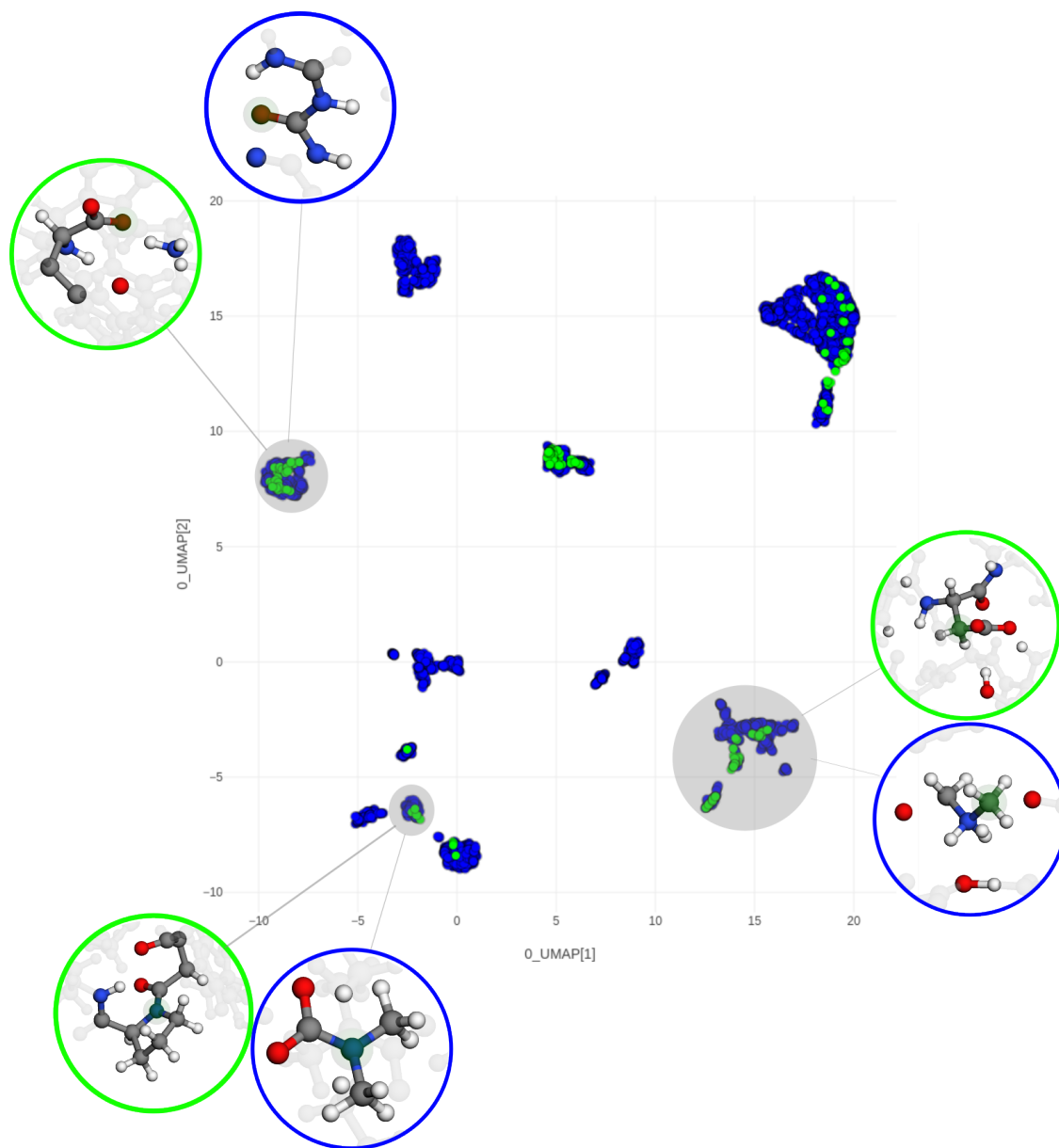


Figure S11: Comparison of the atomic environments in the 99 filtered structure of the training data (blue dots) and in the Chignolin mutant structure (green dots) in the form of a UMAP plot. Insets of some local environments are shown for a few key functional groups of the protein (green circles) with corresponding similar clustered environments from the training set (blue circles)

Fine-tuning

Fine-tuning of the MACE-MP-0b3 model was performed with 100 configurations obtained from the simulations done with the model itself. Specifically, 50 samples of the "Unfolded" Chignolin simulation with 25 in the first third of the simulation where the system is less compact and 25 in the rest of the simulation, 25 of the "Folded" Chignolin simulation and 25 of the "Mutant" Chignolin. For the latter, we only kept into